EE 344 Project Report

1) Title and Team Information

Project Title: Resume Screening and Hiring Prediction System

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Course name: EE 344 — Data Science and Machine Learning

2) Introduction and Problem Description

This project is useful in business analytics and human resource management. Modern companies receive thousands of resumes for each open position, making manual screening slow, inconsistent, and biased. So our project aims to develop a machine learning based system that can automatically assist in resume screening by predicting hiring decisions. From a data science perspective, this task is a binary classification problem, where the model learns from data to predict one of two outcomes: hire or reject.

3) Dataset Description

The dataset is named AI-Powered Resume Screening Dataset, it was downloaded from Kaggle.

Available at:
https://www.kaggle.com/datasets/mdtalhask/ai-powered-resume-screening-dataset-2025

The dataset contains 1,000 samples and 11 features. The dataset is relatively small compared to real-world hiring datasets.

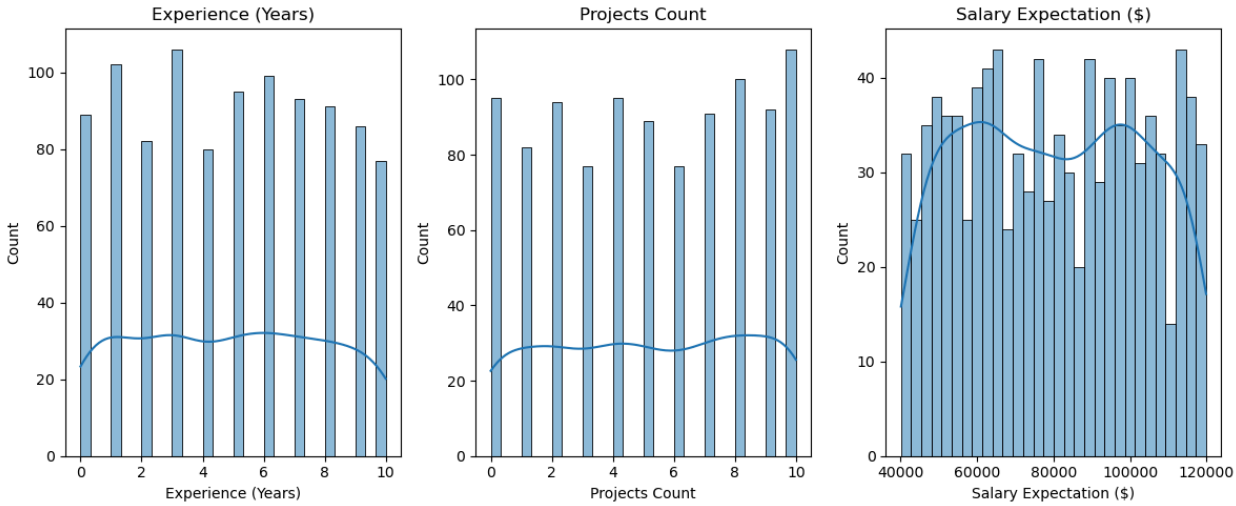
In addition, the dataset has several limitations, including missing values, class imbalance, and noise. The Certifications feature contains missing entries, which were addressed during preprocessing. The Recruiter Decision label is imbalanced.. Furthermore, text-based features such as Skills may introduce noise due to redundant or irrelevant terms, increasing the complexity of the feature space.

Features:	Feature types
Resume_ID: Unique identifier	Removed Irrelevant Identifier Feature
Name: Candidate's name	Removed Irrelevant Identifier Feature
Skills: List of relevant technical skills	Text
Experience (Years): Total work experience	numerical
Education: Highest qualification	Categorical

Certifications: Relevant industry certifications	Text
Job Role: Target job position	Categorical
Recruiter Decision: Hire or Reject	Categorical (Target)
Salary Expectation ($): Expected salary	numerical
Projects Count: Number of projects completed	numerical
AI Score (0-100): AI-based resume ranking score	Numerical (AI Score was dropped)

4) Exploratory Data Analysis (EDA)

The numerical value distribution is not highly skewed.. However, IQR clipping was applied to ensure robust outlier handling.



Irrelevant features such as Resume_ID and Name were dropped. The AI Score is derived from similar features used in this model and is therefore strongly correlated with the hiring decision.. We dropped it to prevent data leakage.

The Certifications feature contains missing values. These were handled by replacing missing entries with an empty string.

The dataset shows a class imbalance, with approximately 80% “Hire” and 20% “Reject” samples. This was addressed using SMOTE

The Skills feature includes multiple skills in a single entry, which may introduce noise and redundancy. This increases the dimensionality of the feature space and may include irrelevant or repetitive terms.

5) Data Preprocessing and Feature Engineering

During Data Cleaning, two irrelevant features were removed. Additionally, the AI Score feature was excluded to prevent data leakage.

The Missing Values was detected on the Certifications feature. This was handled by replacing them with an empty string.

Categorical features such as Education and Job Role were transformed using one-hot encoding.

Text feature Skills are processed using TF-IDF vectorization, and we controlled the number of TF-IDF features to reduce noise.

Numerical features including Experience, Salary Expectation, and Projects Count. THe outlier was handled using IQR clipping and normalized using Min-Max scaling.

The imbalance in the target variable was addressed by SMOTE.

To prevent data leakage, the dataset was split into training and testing sets before applying preprocessing steps.

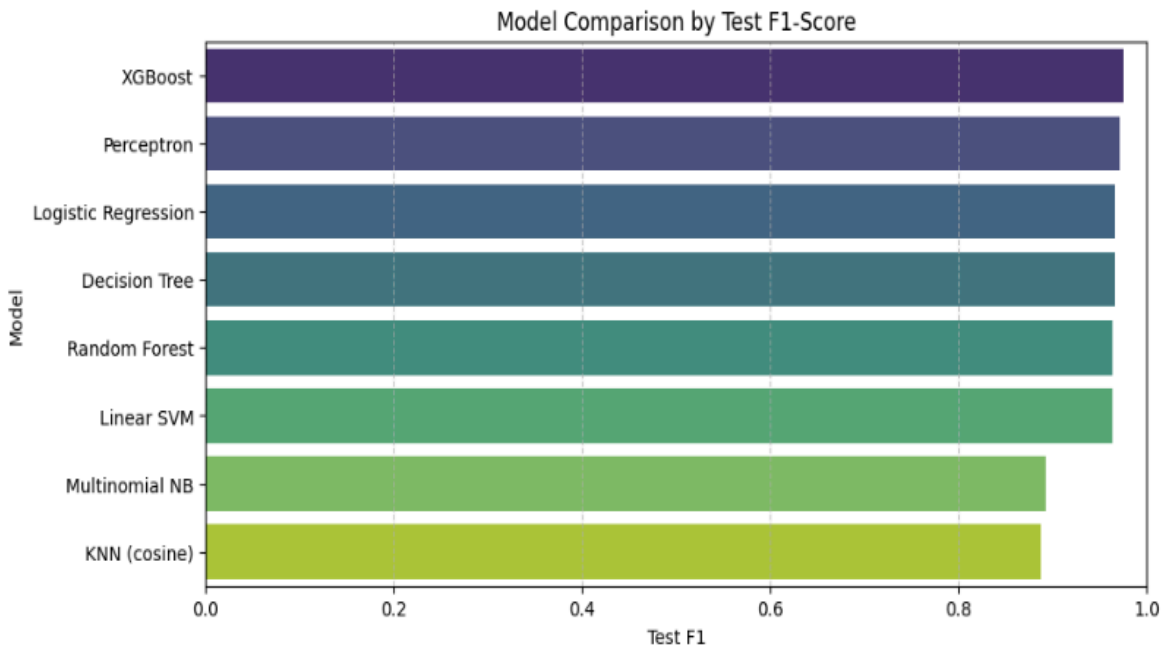
6) Methods / Models

To ensure the model generalizes to real-world resumes, we implemented a pipeline consisting of the following components:

1. Stratified 5-Fold Cross-Validation. Each model was tested on five different subsets of the data to ensure stability and reduce the variance associated with a single train/test split.
2. Resampling using SMOTE. We applied Synthetic Minority Over-sampling Technique(SMOTE) to address class imbalance. Before resampling, we have 650:150 for Hire v.s. Reject, and after resampling, we now have 650:650. This method was applied only to the training portion to prevent data leakage and ensure the validation set remains a real-world distribution.
3. Feature Scaling. We chose MinMaxScaler to normalize numerical features because outliers are removed in our dataset. We also used TF-IDF vectors, ensuring compatibility with distance-based and probabilistic models.

We established a **Logistic Regression** model as our baseline to determine a linear decision boundary for keyword-based classification. We then evaluated several other models to test which model gives the best result.

	Model	CV_F1	CV_PR_AUC	Test Acc	Test Prec	Test Rec	Test F1
0	XGBoost	0.9755	0.9983	0.960	0.9812	0.9691	0.9752
1	Linear SVM	0.9690	0.9965	0.960	0.9873	0.9630	0.9750
2	Random Forest	0.9679	0.9957	0.955	0.9693	0.9753	0.9723
3	Decision Tree	0.9525	0.9440	0.955	0.9935	0.9506	0.9716
4	Logistic Regression	0.9603	0.9972	0.950	0.9935	0.9444	0.9684
5	Perceptron	0.9753	0.9972	0.945	1.0000	0.9321	0.9649
6	Multinomial NB	0.8124	0.9102	0.780	0.8933	0.8272	0.8590
7	KNN (cosine)	0.7471	0.8693	0.655	0.8661	0.6790	0.7612



From the chart, our experimental results indicate that XGBoost achieved the highest performance with a 0.9752 Test F1-Score.

7) Experimental Setup and Evaluation Methodology

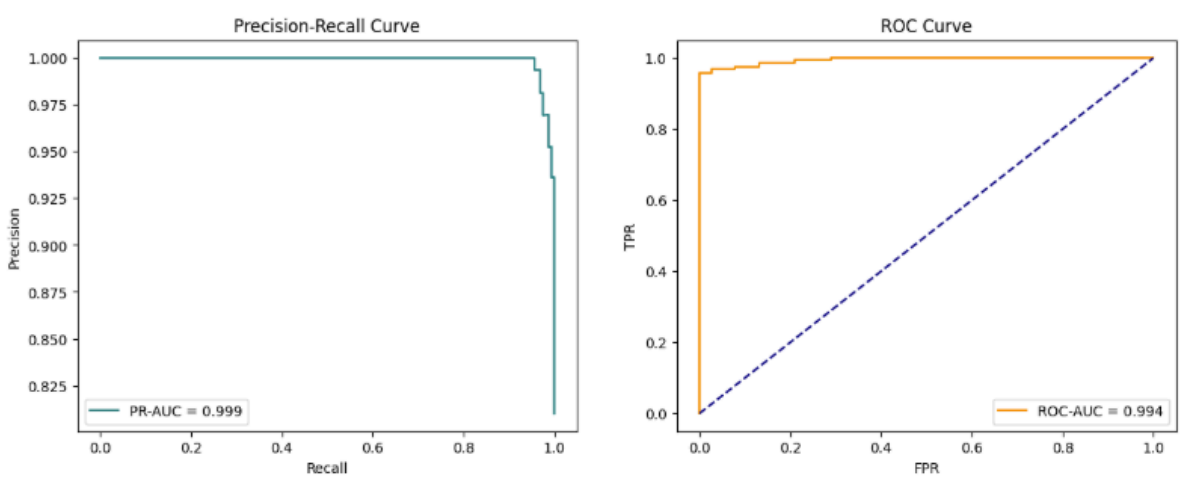
The dataset was initially partitioned into an 80/20 train-test split. To deal with class imbalance, SMOTE was applied to the training set, resulting in a balanced distribution of 650 "Hire" and 650 "Reject" samples.

To provide a comprehensive view of model efficacy, especially given the minority "Reject" class, we moved beyond simple accuracy to prioritize the following metrics:

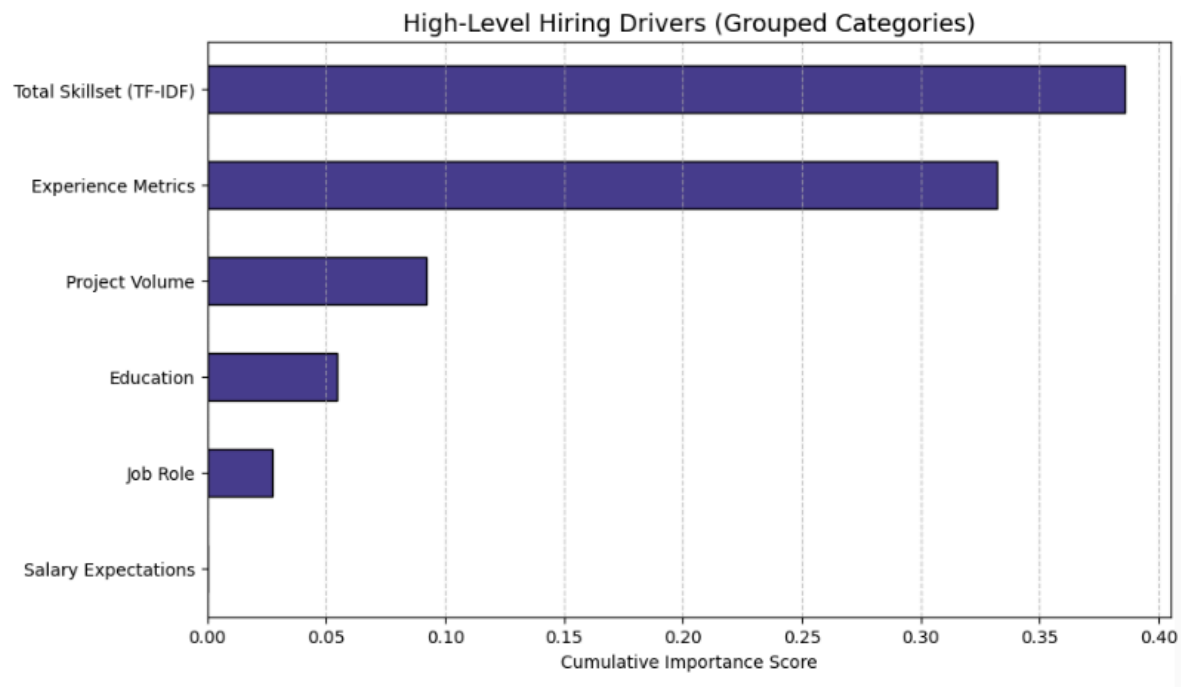
- **F1-Score:** The harmonic mean of precision and recall, serving as our primary ranking metric.
- **Precision-Recall AUC (PR-AUC):** Calculated for our champion model to measure the quality of the "Hire" pool, which is more informative than ROC-AUC for imbalanced datasets.
- **ROC-AUC:** Used to measure the model's overall discriminative ability to separate the two classes across all classification thresholds.
- **Confusion Matrix:** Utilized for our best model to visualize the specific rates of False Positives and False Negatives.

8) Results

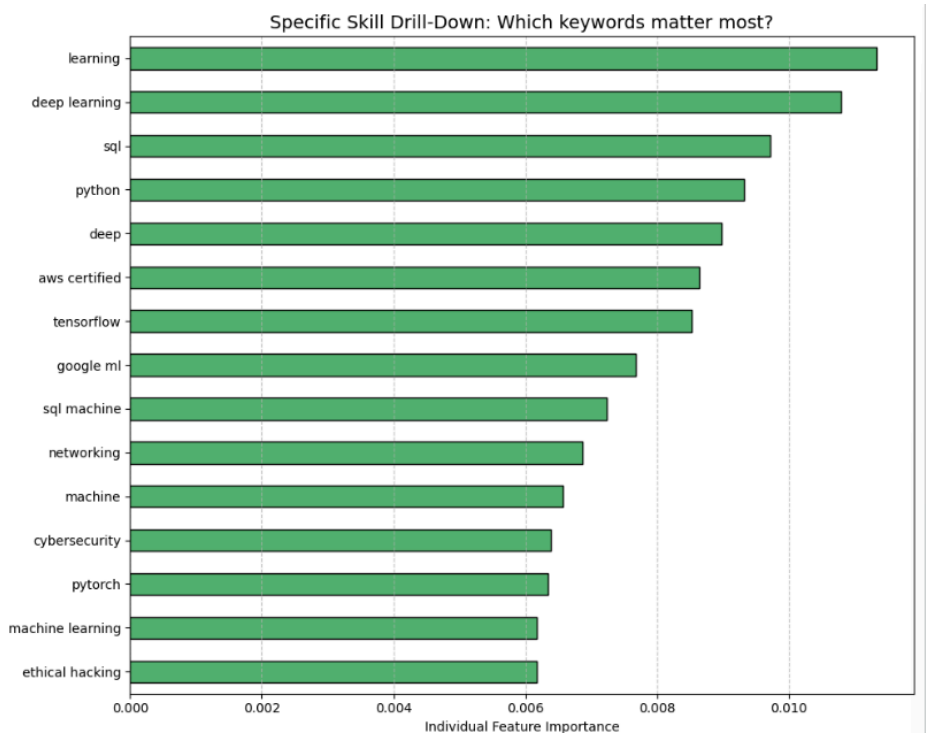
1. Model Performance. XGBoost has the best performance. It achieved a **Test F1-Score of 0.9752**, the area under PR-AUC is 0.9986, and the area under ROC Curve is 0.9940. These near-perfect scores indicate that the model is highly effective at distinguishing between "Hire" and "Reject" candidates within this specific dataset.



2. Feature Importance.
   Our analysis of feature gain a clear hierarchy in hiring drivers. The primary driver is total skillset, and the second and the third are Experience measured by years, and project volume measured by project numbers.



When analyzing and ranking individual technical keywords in skills section using unprocessed and non-resampled data, we are able to say that deep learning is the most important skill and SQL is the second.



3. Ablation study result.
   To understand the contribution of different data types, we compared 3 model configurations, numeric only. Numeric only showed the best F1 score.

	Accuracy	Precision	Recall	F1	Feature Group
0	0.975	0.987578	0.981481	0.984520	Numeric
1	0.735	0.804469	0.888889	0.844575	Text
2	0.965	0.981366	0.975309	0.978328	Combined

9) Discussion and Interpretation

In this section, we interpret the “why” behind our results.

1. XGBoost gives the best performance among all the models with a Test F1 Score of 0.9752. As a gradient boosting framework, it builds tree sequentially where each new tree corrects the errors of the previous ones. It’s specifically designed to handle tabular data(experiences/salary) with sparse features(skills). That might be why it’s more efficient than standard decision trees.
2. The numerical data (like years of experience and project counts) were the most reliable predictors. These are "clean" data points that are easy for the model to understand without confusion.
3. The technical skills section is the most important part of a resume, but using TF-IDF to read it is difficult. Because it treats every word as a separate piece of data, it creates "noise" and high dimensionality. It can't tell that two different words might mean the same thing, which limits its "intelligence."
4. We analyzed specific skills using the raw dataset because SMOTE creates "synthetic" resumes by averaging data points. So the skills will be messed up by random words combinations. While SMOTE helps the model learn, it can blur the importance of specific words. By using the raw data, we ensured our "Top Skills" list represents actual skills found in real resumes.
5. TF-IDF was successful for this project, but a real-world tool would need a "Transformer" model. This would allow the AI to understand the meaning of a sentence rather than just looking for specific keywords.

10) Conclusion and Future Work

This project successfully demonstrated the potential of utilizing machine learning into the resume screening process. We first preprocessed a hybrid dataset of numerical metrics and textual data. Then we applied TF_IDF to convert resume text into a high-dimensional feature space and implemented SMOTED to balance the training set to a 1:1 Hire: Reject ratio. We also utilized stratified 5-fold cross-validation to ensure stability.

XGBoost's performance is the best among all the models with a 0.9752 F1 score, due to its ability to handle the complex relationship between dense numbers and sparse TF-IDF text. While this dataset only included limited samples and job positions, it cannot reflect the whole job market and obviously cannot guarantee the results. But it still provided valuable insights for both recruiters and candidates. Our findings suggest that improving skills and experiences are really important to get outstanded in resume screening. Specifically, for software-related jobs, expertise in deep learning and SQL allows candidates to standout.

In the future, we aim to:

1. Expand the model's training to a larger, more diverse dataset containing a wider variety of industries and real-word data.
2. Transit from classification to ranking to accommodate real-world job markets.
3. Change TF-IDF to transformer. This would allow the system to understand the context and intent of a resume rather than relying on simple keyword matching.
4. Handle bias. Implementing "Fairness-Aware" algorithms to ensure the model does not penalize candidates based on demographic factors or non-traditional career paths.
5. Apply reinforcement learning. We can create a loop where recruiter decisions act as a reward signal. By feeding "corrective" human input back into the model, the AI can continuously refine its understanding of a good fit and adapt to constantly changing market needs over time.

11) Team Contributions

Jiaze Gao	Data & Feature Pipeline - Dataset cleaning and preprocessing. - Feature engineering
Cici Zhang	Modeling & Evaluation - Implementation of baseline models (logistic regression) - Development of advanced models and evaluation matrics - Built a website to simulate the results
Shared	Final Model Analyzation - Model evaluation using classification and regression metrics - Visualization of model performance and feature importance - Final report writing and presentation preparation

12) References

AI-Powered Resume Screening Dataset. Kaggle.
https://www.kaggle.com/datasets/mdtalhask/ai-powered-resume-screening-dataset-2025